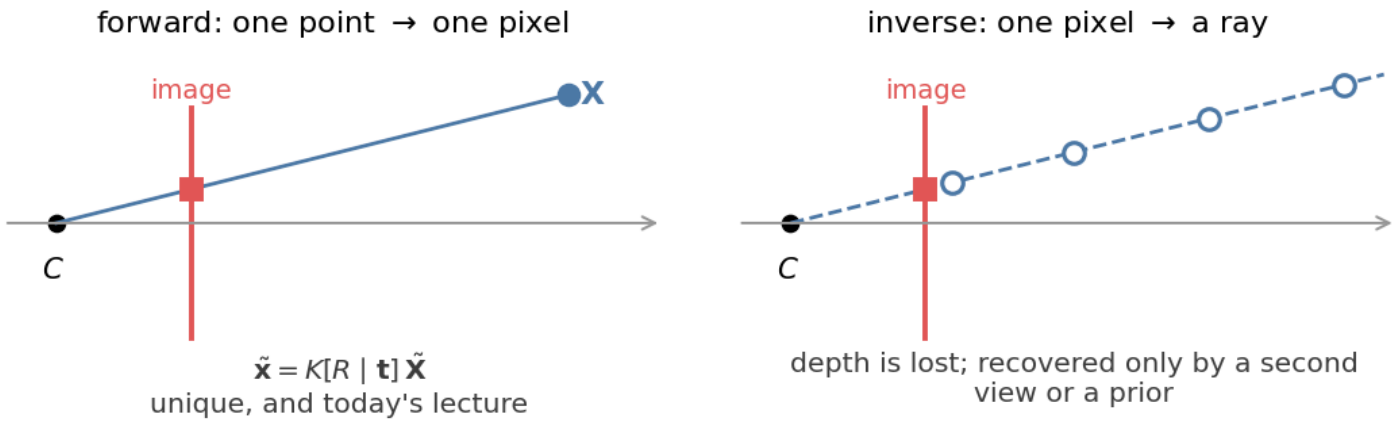
Industrial Vision

Lecture 10 - The camera model

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Where we are

In this lecture, we will look at the image formation.



- Every image is a **projection of a 3D world** onto a plane, and that projection destroys information. Today we write it down. Everything from here to the end of the course is an attempt to invert what we write down today.
- **Today** we examine K and the pose $(R, \mathbf{t})$ in detail. Together they constitute the **forward model**, which answers one question: *given a point in three dimensions, where exactly does it appear in the image?*
- Once the forward map is established the inverse problems become tractable: the homography, calibration, stereo, and structure from motion. The model has two parts: the **intrinsic parameters**, which describe the interior of the camera, such as the focal length, and the **extrinsic parameters**, which describe where the camera is situated in the world.

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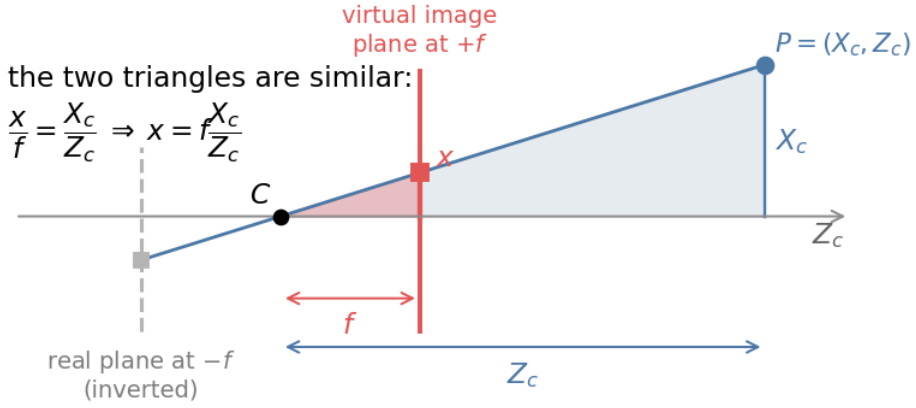
The pinhole camera and perspective projection

Let the optical axis be the Z_c axis, so that the X_cY_c plane is parallel to the image plane, which lies a distance f in front of the optical center. A scene point $P = (X_c, Y_c, Z_c)$ is imaged according to **similar triangles**:

$$\frac{x}{f} = \frac{X_c}{Z_c}, \qquad \frac{y}{f} = \frac{Y_c}{Z_c}$$

$$\boxed{x = f\,\frac{X_c}{Z_c}, \qquad y = f\,\frac{Y_c}{Z_c}}$$

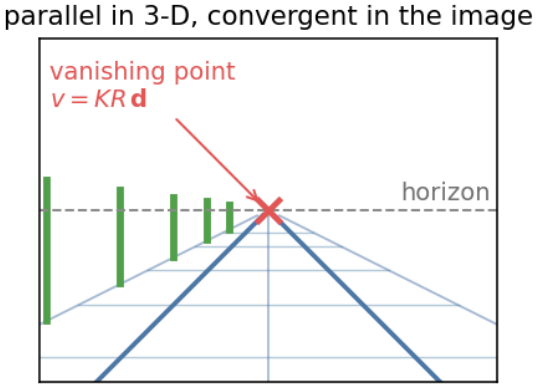
These are the **equations of perspective projection**.

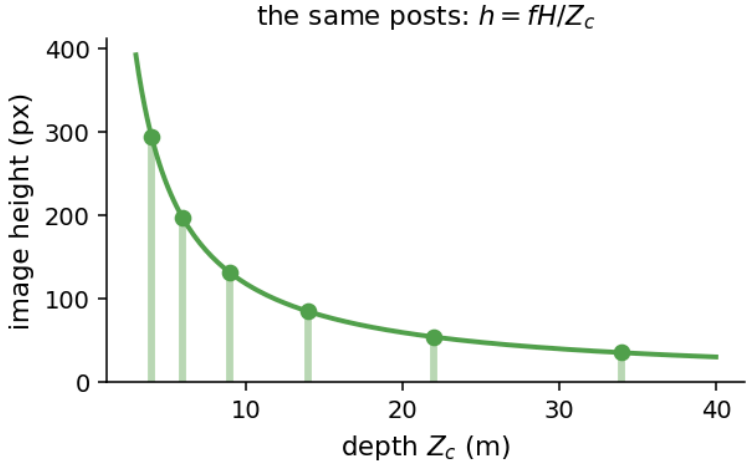


It is the division by Z_c that causes parallel railway tracks to appear to **meet**, since the depth Z_c enters nonlinearly. In order to make the relation linear we pass to **homogeneous coordinates**.

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What the division by Z_c does to the picture





Parallel lines meet. Two rails are two lines $\mathbf{X}(\lambda) = \mathbf{A} + \lambda\mathbf{d}$ sharing the direction $\mathbf{d}$ and differing only in the offset $\mathbf{A}$. Project, then divide through by λ:

$$x(\lambda) = f\,\frac{A_x + \lambda d_x}{A_z + \lambda d_z} = f\,\frac{A_x/\lambda + d_x}{A_z/\lambda + d_z} \xrightarrow[\lambda\to\infty]{} f\,\frac{d_x}{d_z}$$

The offset **divides away** and only the direction survives, so all lines of direction $\mathbf{d}$ tend to one **vanishing point**.

Equal objects shrink as $1/Z$. A post of height H at depth Z_c images at $h = fH/Z_c$, so the product $h\,Z_c$ is fixed.

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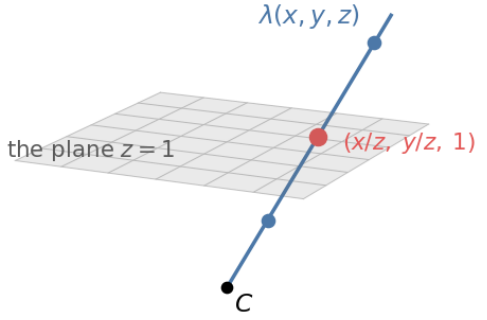
The homogeneous form

We write the image point in homogeneous form as (x, y, z), so that the true pixel coordinates are x/z and y/z. Perspective projection then becomes a **matrix product**:

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \underbrace{\begin{bmatrix} f & 0 & 0 \\ 0 & f & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{K} \underbrace{\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}}_{[I \,|\, \mathbf{0}]} \begin{bmatrix} X_c \\ Y_c \\ Z_c \\ 1 \end{bmatrix} = \begin{bmatrix} fX_c \\ fY_c \\ Z_c \end{bmatrix}$$

Dividing by the last coordinate $z = Z_c$ gives $x/z = fX_c/Z_c$ and $y/z = fY_c/Z_c$, which are exactly the projection equations. The block $[I \mid \mathbf{0}]$ expresses the fact that the point is already given in camera coordinates, and it is replaced by the camera pose below.

This simplest form $K = \mathrm{diag}(f, f, 1)$ assumes that the optical axis passes through the **center** of the image. Manufacturing tolerances make this untrue in practice, and we now correct for it.



One image point is a whole **line through** C. Every $\lambda(x, y, z)$ on it survives the division unchanged, which is what "defined **up to scale**" means

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Four physical corrections that build up K

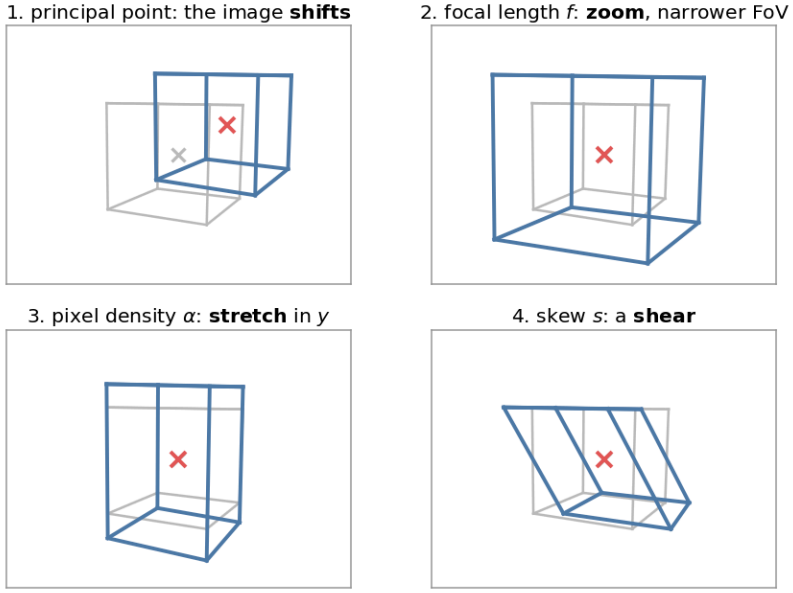
1. Offset of the principal point. The optical axis may meet the sensor at a point displaced from the center by (p_x, p_y), which fills the **third column**: $x = f\frac{X_c}{Z_c} + p_x$.

2. Focal length and field of view. The $(1,1)$ entry sets how many pixels one radian is worth. For a sensor of width w, $\tan\theta = \frac{w/2}{f}$, so $f = \frac{w}{2\tan\theta}$ and $\mathrm{FoV} = 2\theta$.

3. Pixel density, or pixel aspect ratio. Pixels more closely spaced along one axis make the two focal terms differ, so the $(2,2)$ entry becomes αf while $f_x = f$.

4. Skew. Sensor axes that are not exactly orthogonal put a **shear** s in the $(1,2)$ entry. In practice s is very nearly zero for real cameras.

$$K = \begin{bmatrix} f & s & p_x \\ 0 & \alpha f & p_y \\ 0 & 0 & 1 \end{bmatrix}$$

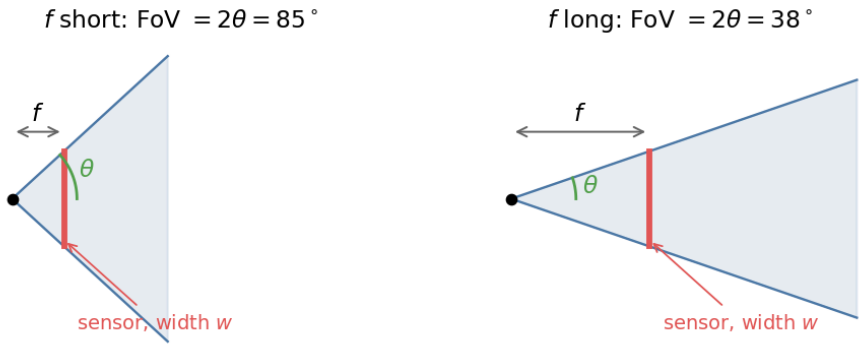


grey: the baseline $K = \mathrm{diag}(f, f, 1)$ red ×: the principal point

One cube, one pose, four different K. Grey is the baseline $K = \mathrm{diag}(f, f, 1)$; the red cross is the principal point. Each correction acts on the **image**, never on the scene.

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The focal length



A focal length in pixels means little by itself. Paired with the sensor width w (in same units) it fixes the angle the camera takes in:

$$\tan\theta = \frac{w/2}{f} \;\Rightarrow\; f = \frac{w}{2\tan\theta}, \qquad \mathrm{FoV} = 2\theta$$

- A **short** f gives a wide field and strong perspective; a **long** f gives a narrow field and a flatter-looking scene.
- Doubling f doubles every offset from the principal point.

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The intrinsic matrix K

$$K = \begin{bmatrix} f & s & p_x \\ 0 & \alpha f & p_y \\ 0 & 0 & 1 \end{bmatrix}$$

5 unknowns: $f,\ \alpha,\ s,\ p_x,\ p_y$

$f:$ focal length $\quad \alpha:$ pixel density

$s:$ skew $\quad (p_x, p_y):$ principal-point offset

- The matrix is **upper-triangular** with a **non-zero diagonal**, and is therefore always **invertible**.
- These parameters are **fixed for a given camera** and are determined once by **calibration**. If calibration is omitted, every subsequent triangulation is biased, because the image point is placed incorrectly.

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The extrinsic parameters relate the world to the camera

A point is generally given in a **world frame** (X_w, Y_w, Z_w) rather than in the camera frame, because a second camera, or a calibration rig, defines its own reference frame. Let C denote the camera center expressed in world coordinates.

The point is first translated and then rotated:

$$\mathbf{X}_c = R\,(\mathbf{X}_w - C)$$

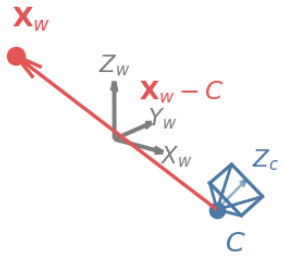
- Subtracting C moves the origin to the camera center.
- Applying R aligns the axes with those of the camera.

In homogeneous form this is a single 4×4 matrix:

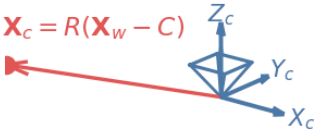
$$\tilde{\mathbf{X}}_c = \begin{bmatrix} R & -RC \\ \mathbf{0}^\top & 1 \end{bmatrix} \tilde{\mathbf{X}}_w$$

R has three degrees of freedom and C three, giving **six extrinsic degrees of freedom**.

1. subtract C
the origin moves to the camera



2. apply R
the axes align with the camera



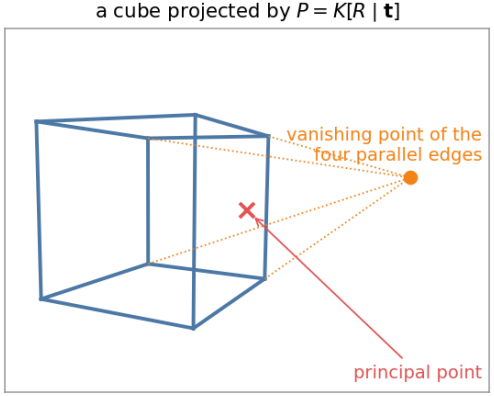
The camera projection matrix P

Composing the intrinsic and extrinsic transformations maps a **world** point in three dimensions to its image:

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = K\,[\,I \mid \mathbf{0}\,] \begin{bmatrix} R & -RC \\ \mathbf{0}^\top & 1 \end{bmatrix} \tilde{\mathbf{X}}_w = \underbrace{K\,[\,R \mid -RC\,]}_{P}\ \tilde{\mathbf{X}}_w$$

$$\boxed{P = K\,[\,R \mid \mathbf{t}\,], \qquad \mathbf{t} = -RC} \quad \text{a } 3 \times 4 \text{ matrix}$$

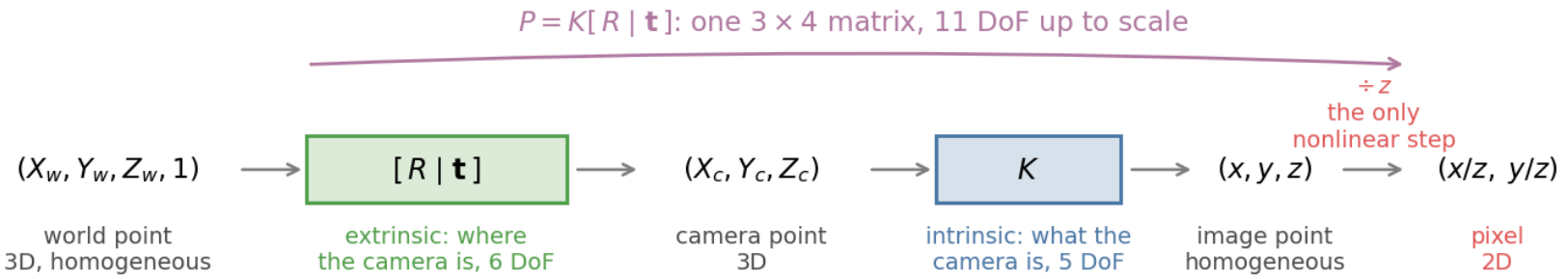
There are two equivalent readings. The form $[R \mid \mathbf{t}]$ corresponds to rotating and then translating, whereas $KR[\,I \mid -C\,]$ corresponds to translating and then rotating. Both are correct, and $[R \mid \mathbf{t}]$ is the standard convention.



A cube put through this very P. Four edges that are parallel in three dimensions meet at one image point, and the red cross marks where the optical axis pierces the sensor.

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The whole forward model in one picture

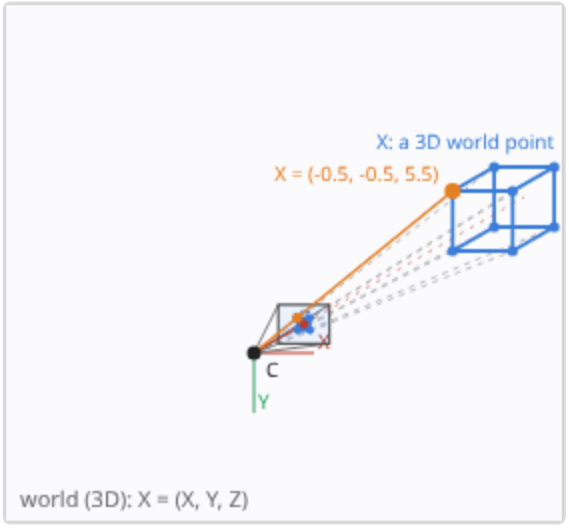


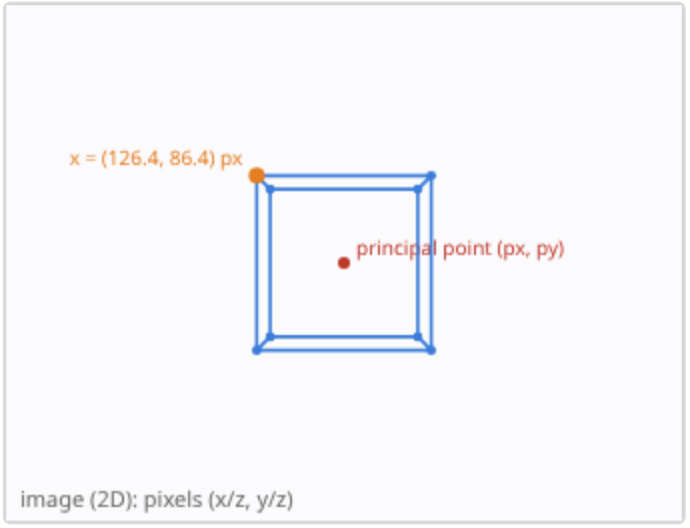
- The matrix P has twelve entries but is defined only **up to scale**, since P and λP produce identical images. It therefore has **eleven** unknowns, comprising five from K and six from (R, C).
- Only the last step is nonlinear. Everything before it is one matrix product, which is the entire reason for working in homogeneous coordinates.
- It is important to note that P **maps three-dimensional points to the image**, whereas a **homography maps one image to another**. The two are different objects.

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Inline demonstration

Left: a 3D cube, the camera centre **C**, and the image plane; rays from **C** through the cube pierce the plane at its image. Right: that image as pixels, **x ~ K[R | t] X**. The orange corner is followed from its world coordinates **X** to its pixel **x**. The five entries of **K** (f, α, s, p_x, p_y) are intrinsic and reshape the sensor; **yaw** and **shift** are extrinsic and move the camera.





focal length f = 480 px

pixel density α = 1.00 (αf = 480 px)

skew s = 0 px

camera yaw = 0°

principal pt p_x = 170 (offset +0)

principal pt p_y = 130 (offset +0)

camera shift C_x = 0.00

```
K (intrinsic)   [R | t] (extrinsic)
 480    0  170    1.00  0.00  0.00 |  0.00
   0  480  130    0.00  1.00  0.00 |  0.00
   0    0    1    0.00  0.00  1.00 |  0.00
```

FoV 2·atan(w/2f) = 39.0°

The five sliders that edit *K*, namely *f*, *α*, *s*, p_x and p_y, are **intrinsic**: they reshape the sensor and move nothing in the world. The yaw and the shift are **extrinsic**: they move the camera and leave *K* untouched.

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A rotation is not nine numbers

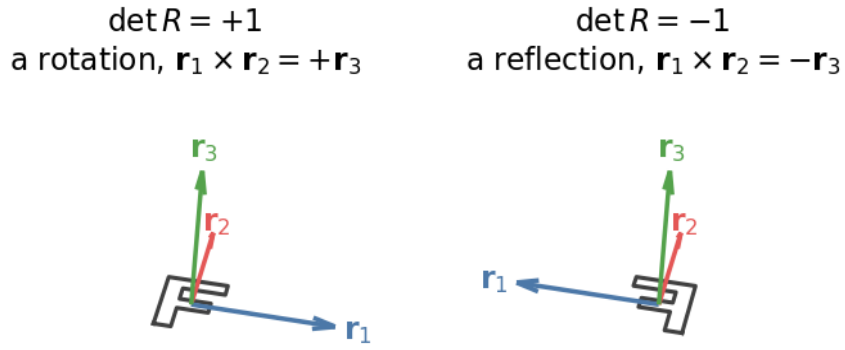
The matrix R has nine entries, but it may not take nine arbitrary values. Its columns are the axes of the camera frame written in world coordinates, and they are unit vectors at right angles to one another:

$$R^\top R = I, \qquad \det R = +1$$

The first condition is **six scalar equations**: three unit-norm conditions, one for each column, and three orthogonality conditions, one for each pair. Nine entries less six equations leave $9 - 6 = 3$ **degrees of freedom**.

The second is not a seventh equation but a **selection**: $R^\top R = I$ also admits $\det R = -1$, and those matrices are **reflections**, which reverse handedness.

A pose therefore has **six degrees of freedom**, three for R and three for $\mathbf{t}$, though $[\, R \mid \mathbf{t} \,]$ has twelve numbers.

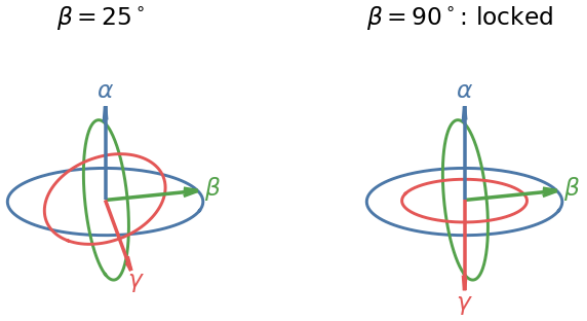


The columns of R form an orthonormal frame; the determinant fixes its handedness. The right-hand glyph is the mirror image of the left.

Three ways to write a rotation (Hartley and Zisserman, Ch. 6)

Representation	Numbers	Constraints	Composition	Singularities
Rotation matrix R	9	6	matrix product	none
Euler angles (α, β, γ)	3	0	awkward	**gimbal lock**
Axis and angle $\boldsymbol{\omega} = \theta\hat{\mathbf{n}}$	3	0	awkward	at $\theta = 0$ and $\theta = \pi$
Unit quaternion q	4	1	quaternion product	none, but a double cover

No representation is at once **minimal**, **unconstrained** and **free of singularities**. That is a property of the set of rotations itself rather than a defect of notation.



Gimbal lock in the Z-Y-X convention. At $\beta = 90°$ the γ axis has swung onto the α axis, so two turns act on the same axis.

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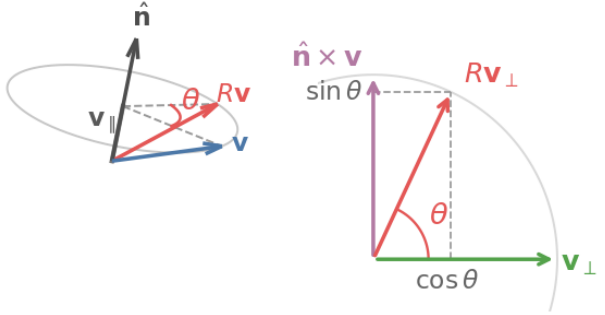
Axis and angle, and Rodrigues' formula

By **Euler's rotation theorem**, every rotation about a fixed origin is a turn by one angle θ about one unit axis $\hat{\mathbf{n}}$, written as one vector $\boldsymbol{\omega} = \theta\hat{\mathbf{n}}$.

The turn uses the **skew-symmetric matrix** of the axis:

$$[\hat{\mathbf{n}}]_\times = \begin{bmatrix} 0 & -n_3 & n_2 \\ n_3 & 0 & -n_1 \\ -n_2 & n_1 & 0 \end{bmatrix}, \qquad [\hat{\mathbf{n}}]_\times \mathbf{v} = \hat{\mathbf{n}} \times \mathbf{v}$$

This operator returns twice: L18 builds the fundamental and essential matrices from $[\mathbf{t}]_\times$, and L16 the exponential map from $[\boldsymbol{\omega}]_\times$.



Left, the part of $\mathbf{v}$ along the axis is untouched and the perpendicular part turns by θ. Right, the same turn seen face on in that plane.

The inversion below fails at $\theta = 0$, where every axis will serve, and at $\theta = \pi$, where $R - R^\top = \mathbf{0}$ and the sign of the axis is lost.

Rodrigues' formula assembles the matrix from the axis and the angle:

$$\boxed{R = I + \sin\theta\,[\hat{\mathbf{n}}]_\times + (1 - \cos\theta)\,[\hat{\mathbf{n}}]_\times^2}$$

It inverts, recovering the axis and the angle from a given R:

$$\theta = \arccos\left(\frac{\operatorname{tr} R - 1}{2}\right), \qquad [\hat{\mathbf{n}}]_\times = \frac{R - R^\top}{2\sin\theta}$$

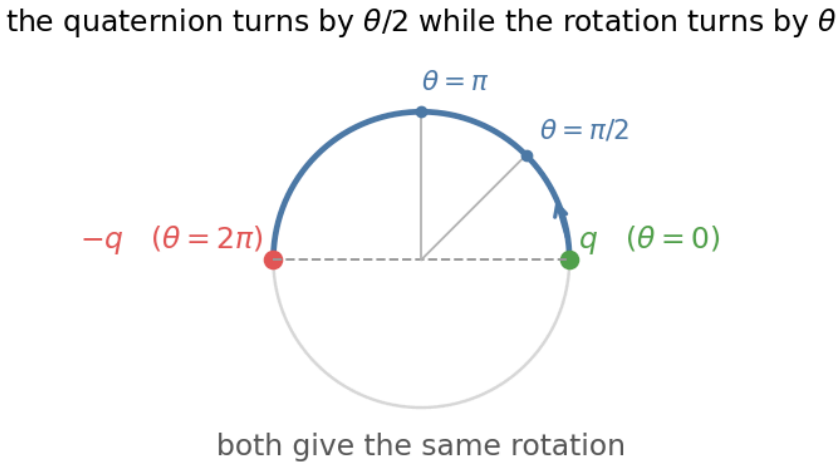
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Unit quaternions

A **unit quaternion** carries the same information with no singularity at all:

$$q = \left(\cos\tfrac{\theta}{2},\ \sin\tfrac{\theta}{2}\,\hat{\mathbf{n}}\right), \qquad \|q\| = 1$$

- Rotations are composed by the **quaternion product**.
- The norm must be one, so an unconstrained step leaves the sphere and every update has to be **renormalised**.
- The rotation is defined for every q, so there is no singularity.
- q and $-q$ describe the **same** rotation, which is called the **double cover**.



The half-angle is the reason for the double cover. As the rotation turns through 2π the quaternion travels only half of its own circle, arriving at $-q$ rather than at q.

Each representation trades one inconvenience for another. We will show how to differentiate with respect to a rotation without accepting any of the trades, by parametrising **locally**, in the tangent space at the current estimate, rather than globally.

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Questions

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Summary

- **Perspective projection** is $x = fX_c/Z_c$ and $y = fY_c/Z_c$. The division by depth accounts for the characteristic effects, including the convergence of parallel lines.
- The **intrinsic matrix** $K = \begin{bmatrix} f & s & p_x \\ 0 & \alpha f & p_y \\ 0 & 0 & 1 \end{bmatrix}$ holds the focal length, the skew, the aspect ratio and the principal point. It has five degrees of freedom, is upper-triangular and **invertible**, and is determined by calibration.
- The **extrinsic parameters** appear in $\mathbf{X}_c = R(\mathbf{X}_w - C)$ and describe the pose of the camera in the world.
- A **rotation** obeys $R^\top R = I$ and $\det R = +1$, so it has three degrees of freedom and a pose has six. It may be written as a matrix, as Euler angles, as an axis and an angle through Rodrigues' formula, or as a unit quaternion, and **no one of these is at once minimal, unconstrained and free of singularities**.
- The **projection matrix** is $P = K[R \mid \mathbf{t}]$ with $\mathbf{t} = -RC$. It is a 3×4 map from **world** points in three dimensions to image points, with eleven degrees of freedom up to scale. It is not a homography.

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References

1. R. Hartley and A. Zisserman, *Multiple View Geometry in Computer Vision*, 2nd ed. Cambridge Univ. Press, 2004. See **Ch. 6, "Camera Models"**

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